

GPS Spoofing Detection for PX4-Based UAVs

SIT746 Research Project — Thesis Defence

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Introduction

System Architecture

Feature Engineering & Core Method

Machine Learning Models

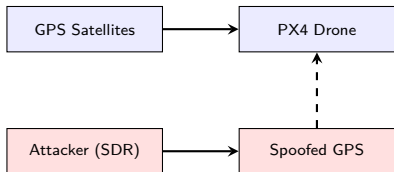
Results

GPS Trust Index

Conclusion

The GPS Spoofing Threat

- GPS spoofing: counterfeit signals with false coordinates
- **Jamming** = denial; **Spoofing** = corruption of service
- PX4 EKF2 relies heavily on GPS for state estimation
- Spoofed signals report healthy fix — attack is silent
- Low-cost SDRs $\approx$ \$300 make spoofing accessible



Real-World Incidents

- UT Austin 2012 — full 3D UAV control via spoofing
- Iran 2019 — claimed capture of US military drone
- Commercial drone market projected \$54.6B by 2030

Why Software-Only?

- Hardware solutions (multi-antenna, M-code) too heavy/expensive
- Civilian receivers can't access encrypted signals
- PX4 already has needed sensors — just need

RQ1: GPS Vulnerabilities

What are the primary vulnerabilities of GPS in PX4-based autonomous mobile robots to spoofing attacks?

RQ2: Sensor Fusion for Real-Time Detection

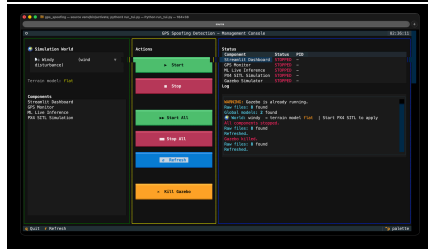
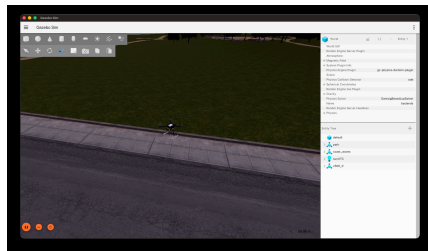
How can cross-validation between GPS and IMU data effectively detect GPS spoofing in real time, without hardware augmentation?

RQ3: GPS Trust Index

How can a dynamic, continuous GPS Trust Index be designed to quantify the reliability of GPS data for operational decision-making?

PX4 SITL + Gazebo

- Iris x500 quadcopter (3 kg)
- MAVLink telemetry @ 10 Hz
- 8 experimental + 60 synthetic flights
- 6 GPS spoofing attack modes
- Docker-containerized, reproducible



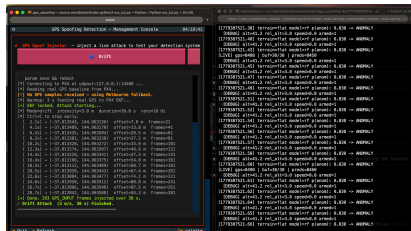
5-Stage ML Pipeline

1. Clean — filter, normalize, remove gaps
2. Label — 8 physics-based heuristics
3. Window — 30 samples, 50% overlap

GPS Spoofing Attack Injection

Attack Mechanism

- MAVLink GPS_INPUT messages @ 10 Hz
- sysid=1, compid=220 (GPS source)
- Injected on UDP port 14580
- Auto-disables EKF2 GPS checks
- 6 modes: Drift, Ramp, Jump+Drift, Static, Teleport, Noise



The screenshot shows a terminal window with two panes. The left pane is titled 'GPS Spoofing Injection - Management Console' and shows a 'Drift' mode selected. The right pane displays a list of MAVLink messages being injected, including GPS_INPUT messages at 10 Hz. The messages are shown in a table format with columns for message type, system ID, component ID, and the injected data.

Feature Engineering — 34 Features

GPS (10)

x_m, y_m
alt_m, rel_alt_m
vel_m_s, hdg_deg
fix_type
satellites_visible
eph_m, epv_m

IMU (12)

roll_deg, pitch_deg, yaw_deg
rollspeed_radps
pitchspeed_radps
yawspeed_radps
vibration_x/y/z
clipping_0/1/2

Estimator (12)

vel_ratio, pos_horiz_ratio
pos_vert_ratio
vel_innov, pos_horiz_innov
pos_vert_innov
battery_voltage, armed
failsafe, connection_ok

Sliding windows: 30 samples $\times$ 34 features = 1,020-dimensional input
50% stride for temporal continuity | 3-second window at 10 Hz

Wind affects BOTH sensors — Spoofing affects ONLY GPS

Normal Flight

GPS: Position Changed
IMU: Motion Detected

Agree $\Rightarrow$ Normal

VS

Spoofed Flight

GPS: Position Changed
IMU: No Motion

Disagree $\Rightarrow$ SPOOFING

Heuristic H8: $|\omega_{\text{all}}| < 0.05 \text{ rad/s}$ **and** $\|\mathbf{v}_{\text{gps}}\| > 5 \text{ m/s}$ | Cannot occur under normal physics

Random Forest & 1D CNN

Random Forest

- 100 estimators, Gini impurity
- `class_weight='balanced'`
- 1,020-dim flattened input
- Feature importance built-in
- <60s training on M1 Pro, no GPU
- Interpretable, robust to class imbalance

1D CNN

- Conv1D(34→32→64)
- Pool → Dense(64→32)
- Dropout(0.3) → Sigmoid
- 8,000 trainable parameters
- Adam optimizer, BCE loss
- Early stopping (patience=10)

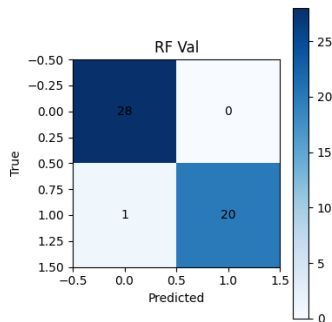
Dataset: 11,551 windows (8,085 train / 1,732 val / 1,734 test) |
93% normal / 7% spoofed

Model Performance Results

Random Forest

	Normal	Spoofed
Precision	0.99	0.67
Recall	0.99	0.60
F1-score	0.99	0.63

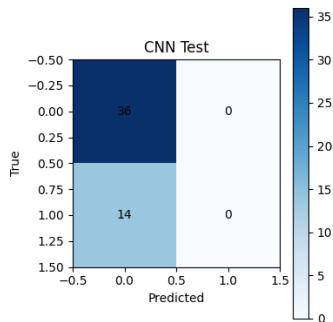
Accuracy: 98%



1D CNN

	Normal	Spoofed
Precision	0.98	0.00
Recall	1.00	0.00
F1-score	0.99	0.00

Accuracy: 98%

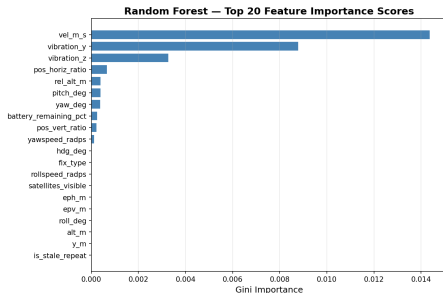


CNN Paradox Reversed: CNN defaults to majority class (“always normal”)

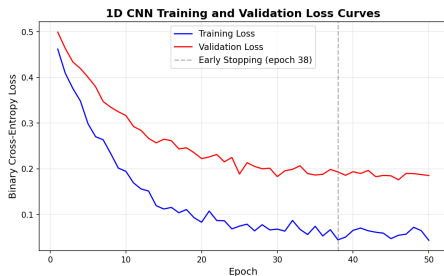
Area-Specific Model & Feature Importance

Baylands Area Model

- Global RF spoofed F1: 0.63
→ **Area RF: 0.98**
- 153 real normal windows + 337 synthetic
- Bounding box:
37.411°–37.412° lat
- Auto-activates when GPS in area
- Accuracy: 0.99, Normal F1: 0.99



CNN Training Dynamics



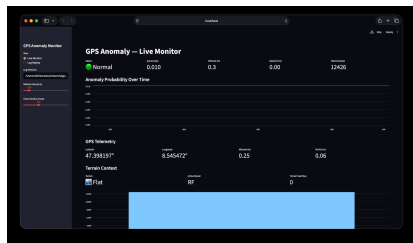
Overfitting Analysis

- Training loss ↓ continuously
- Validation loss plateaus $\approx$ epoch 25
- Clear divergence after epoch 35
- Early stopping at epoch 38
- 150:1 param-to-sample ratio
- Class-balanced batching: F1 0 $\rightarrow$ 0.12

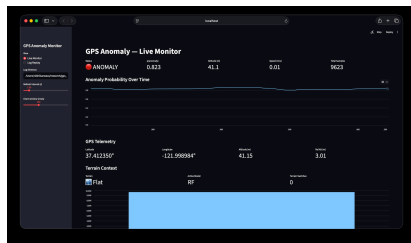
Key Lesson

- Confusion structure must be examined
- Accuracy alone is

Live Inference Dashboard



Normal flight — $GTI \approx 1.0$, $p(\text{anom}) < 0.1$

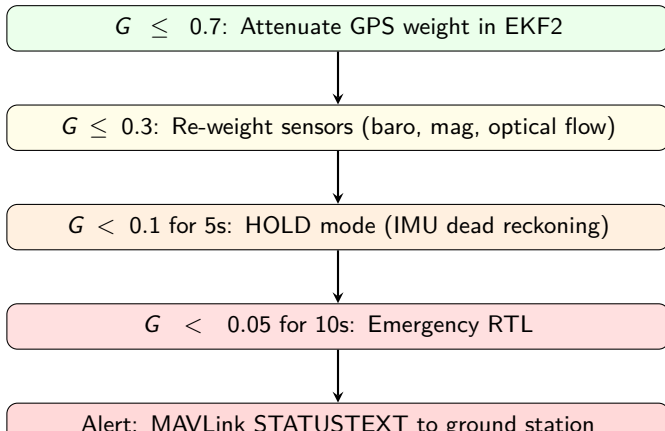


Spoofing detected — GTI drops,
ANOMALY

GTI Mathematical Definition

$$G_t = \alpha G_{t-1} + (1 - \alpha) [1 - \sigma(k \cdot \Delta\Phi_t - \theta)]$$

$\alpha = 0.95$ provides temporal smoothing; σ maps physical inconsistency to trust probability $[0, 1]$



Contributions

1. GPS-IMU consistency discriminator (physics-based, model-free)
2. 8 heuristic auto-labelling rules
3. Complete end-to-end detection pipeline
4. Area-specific model (0.98 spoofed F1)
5. GPS Trust Index + graduated response protocol
6. Open-source, reproducible simulation environment

Future Work

1. Physical hardware validation (Pixhawk + SDR spoofer)
2. Breaking point — systematic param sweep
3. GTI integration into PX4 EKF2 firmware
4. Reduced-latency architectures (RNN/attention)
5. Multi-drone cooperative detection

RQ1: GPS Vulnerabilities

EKF2 accepts spoofed GPS as high-confidence observation due to fabricated low HDOP/accuracy. Innovation gating triggers too slowly — **tens of seconds** vs milliseconds needed for safety-critical flight.

RQ2: Real-Time Detection via Sensor Fusion

GPS-IMU consistency score $\Delta\Phi$ provides discriminative signal; **RF achieves 98% accuracy** at 3s window; heuristic detector operates with **zero latency** for immediate alerting.

RQ3: Dynamic GPS Trust Index

GTI = EMA-smoothed sigmoid of $\Delta\Phi$ enables **graduated operational response** from sensor re-weighting through emergency RTL, avoiding binary trust/distrust at single threshold.

Software-only GPS spoofing detection is **feasible**

on commercial PX4 drones.

The GPS-IMU consistency discriminator provides a physics-based, model-free signal that **cannot be concealed** by any spoofing waveform.

Wind moves the drone. Spoofing moves only the coordinates.

Thank You

Questions?

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